M&M'S: MIX AND MATCH ATTACKS ON SCHNORR-TYPE BLIND SIGNATURES WITH REPETITION

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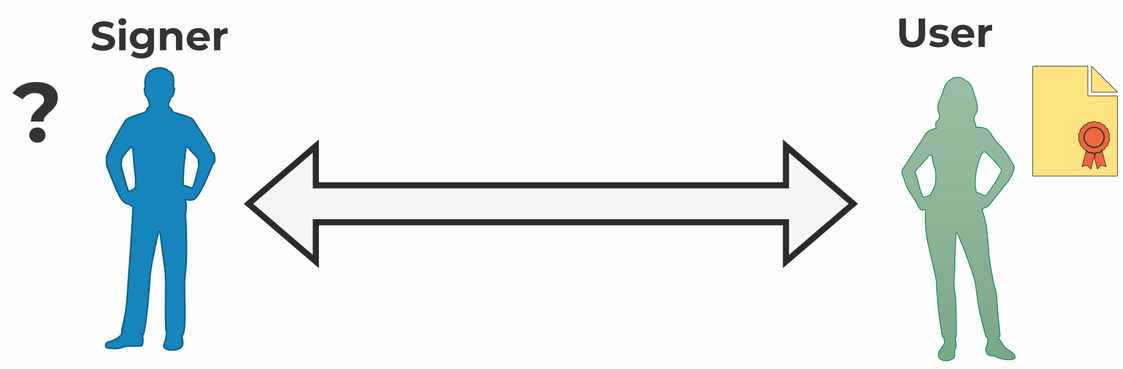
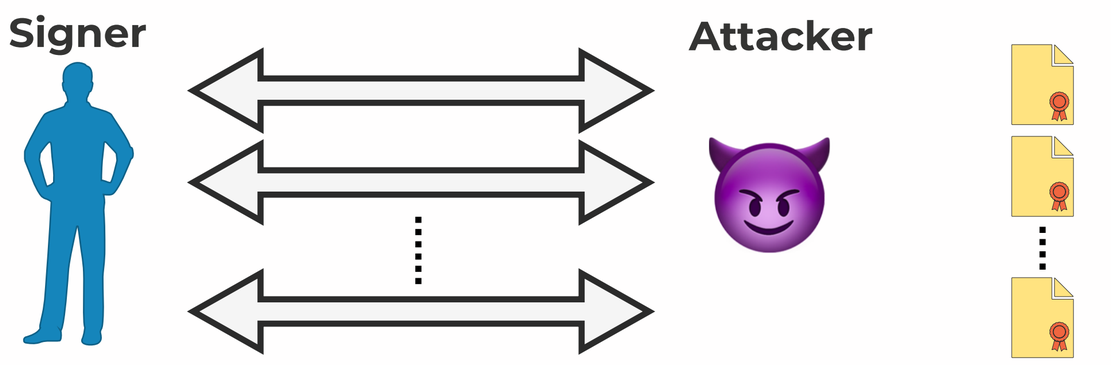
BLIND SIGNATURES



Blindness: the signer does not learn the message
Unforgeability: only one signature with one interaction

CONTRIBUTION

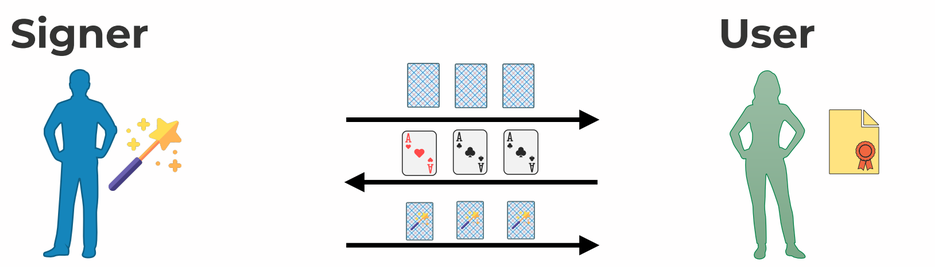


We break unforgeability of Schnorr-type blind signatures with repetitions

Forgery

SCHNORR-TYPE BLIND SIGNATURES WITH REPETITION

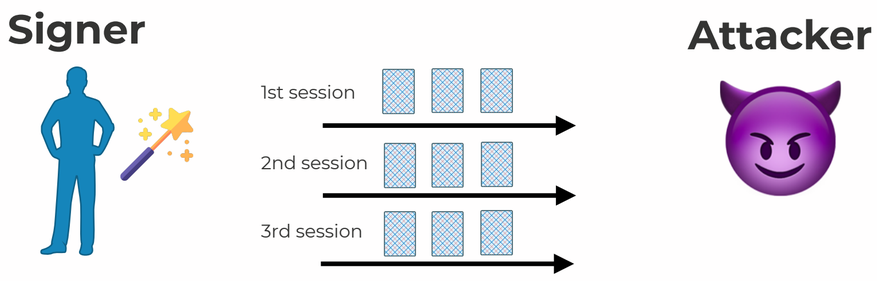
Repeat the protocol in parallel to increase soundness



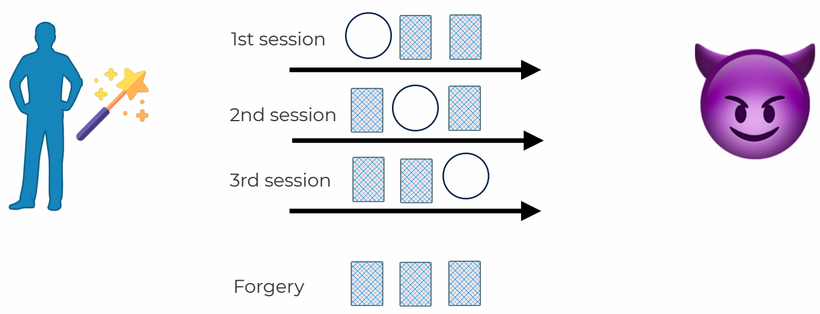
The challenge is derived from the message as the output of an hash function $\mathscr{H}$

MIX AND MATCH

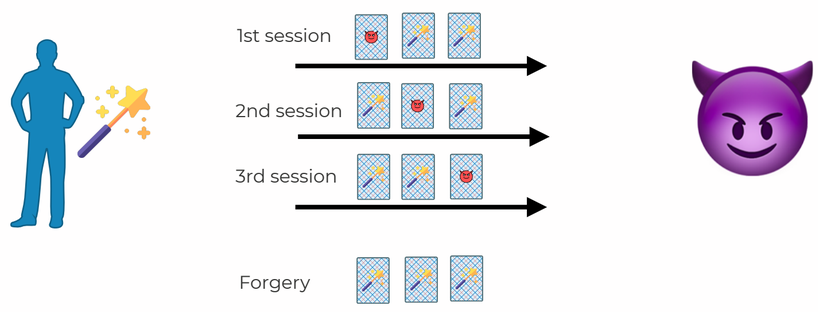
For simplicity, we consider three sessions



1. The attacker opens the sessions in parallel



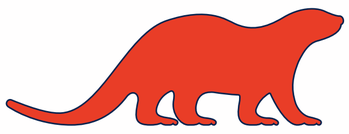
2. The attacker **mixes** the commitments together to craft the forgery



3. The attacker replaces the commitments with some malicious ones and **matches** the corresponding challenges with the values of $\mathscr{H}$

IMPLICATIONS

Impossibility result: Schnorr-type blind signatures with repetitions of a small challenge space **are not concurrently secure**



Affected scheme: CSI-Otter [KLLQ'23] (Crypto 23)
This attack is able to forge **129 valid signatures** with **128 concurrent sessions**

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This poster is based on the following publication:
Do, K., Hanzlik, L., Paracucchi, E. (2024). M&M'S: Mix and Match Attacks on Schnorr-Type Blind Signatures with Repetition. In: Joye, M., Leander, G. (eds) Advances in Cryptology – **EUROCRYPT 2024**